

A Sharpened Asplund Averaging Exponent for Simultaneous AUS/AUC Renormings

Automated Math Discovery Smoke Test

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Source and Open Question

This packet concerns Remark 4.4 from:

Stephen J. Dilworth, Denka Kutzarova, Gilles Lancien, Lovasoa N. Randrianarivony,
Equivalent norms with the property (β) of Rolewicz, arXiv:1506.07978.

The source paper proves that if a reflexive Banach space admits separately an equivalent AUS norm of power type p and an equivalent AUC norm of power type q , then it admits a single equivalent norm which is simultaneously AUS of power type p and AUC of power type $4q$. Immediately after this lemma, the authors ask whether both original power types can be preserved.

The source location is Remark 4.4 on page 10.

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Note that $n_0 \leq \alpha t^{-q} + 1 \leq t^{-q}(\alpha + 1)$. Then, summing equations (4.2), (4.3) and (4.4) we obtain that for all $y \in B_E$ with $|y| \geq t$

$$|x + y| \geq |x| + \frac{D}{2n_0^3} t^q \geq 1 + \frac{D}{2(\alpha + 1)^3} t^{4q},$$

which concludes our proof. $\square$

Remark 4.4. *It follows from Corollary 2.4 that we could also build an equivalent norm which is AUC of power type q and AUS with a power type in $(1, p)$. We do not know if it is possible to preserve both power types together. To the best of our knowledge the similar question for uniformly smooth and convex norms is still open.*

Remark 4.5. *It is worth mentioning that one can use a nice Baire category argument to show that if a reflexive Banach space X admits an equivalent AUS norm and an equivalent AUC norm then it admits an equivalent norm which is simultaneously AUS and AUC. However, the very nature of Baire's Lemma makes it quite hopeless to get a quantitative result such as Lemma 4.3 so easily. Let us nevertheless describe this qualitative argument.*

Proof. We shall adopt here the presentation of the Asplund averaging technique, based on Baire's Lemma that can be found in [4] (p. 52-56). Let us denote by P the set of norms on X that are equivalent to the original norm $\|\cdot\|$ and let B be the closed unit ball of $\|\cdot\|$. For $N, M \in P$, we set

Proof Intuition

Lemma 4.3 in the source paper uses dual norms

$$\|x^*\|_{n,*} = N^*(x^*) + n^{-1}M^*(x^*)$$

and averages their predual norms with weights n^{-3} . In the AUC part of the proof, the paper records a strict t^q gain at one index $n_0 \simeq t^{-q}$, producing the displayed loss t^{4q} .

The same argument actually gives a strict t^q gain for every tail index $n \geq n_0$, not just for the single index n_0 . If the averaging weights are changed from n^{-3} to $n^{-\theta}$ with $\theta > 2$, the AUS estimate is unchanged because $\sum n^{1-\theta} < \infty$. The AUC estimate gains the whole tail

$$t^q \sum_{n \geq n_0} n^{-\theta} \simeq t^q n_0^{1-\theta} \simeq t^{q\theta}.$$

Choosing $2 < \theta < r/q$ gives AUC power type r for every $r > 2q$.

This still does not answer the exact preservation question: the method cannot reach q , and even this weighted version stops strictly above $2q$.

Candidate Partial Result

Theorem 1. *Let X be a reflexive Banach space. Suppose that X admits an equivalent AUS norm of power type p and an equivalent AUC norm of power type q , where $1 < p, q < \infty$. Then, for every $r > 2q$, X admits an equivalent norm which is simultaneously AUS of power type p and AUC of power type r .*

Proof. We modify the proof of Lemma 4.3 in Dilworth–Kutzarova–Lancien–Randrianarivony. Let N be an equivalent norm with

$$\bar{\delta}_N(t) \geq at^q \quad (0 < t \leq 1)$$

and let M be an equivalent norm with

$$\bar{\rho}_M(t) \leq bt^p \quad (0 < t \leq 1).$$

After replacing constants, assume

$$N(x) \leq M(x) \leq CN(x) \quad (x \in X).$$

For $n \geq 1$ define an equivalent dual norm on X^* by

$$\|x^*\|_{n,*} = N^*(x^*) + n^{-1}M^*(x^*),$$

and let $\|\cdot\|_n$ denote its predual norm on X . As in Lemma 4.3 of the source paper, there is a constant K such that

$$\bar{\rho}_{\|\cdot\|_n}(t) \leq Knt^p \quad (0 < t \leq 1, n \geq 1). \quad (1)$$

Also

$$(1 + n^{-1})^{-1}N(x) \leq \|x\|_n \leq N(x) \quad (x \in X). \quad (2)$$

Fix $r > 2q$, and choose θ with

$$2 < \theta < r/q.$$

Define

$$|x| = \sum_{n=1}^{\infty} n^{-\theta} \|x\|_n.$$

By (2), this is an equivalent norm on X .

AUS estimate

Let $|x| = 1$ and $0 < t \leq 1$. Since $|\cdot|$, N , and the norms $\|\cdot\|_n$ are uniformly equivalent, there are constants $A, B > 0$ such that

$$A|z| \leq \|z\|_n \leq B|z|, \quad A|z| \leq N(z) \leq B|z|$$

for all $z \in X$ and $n \geq 1$.

Choose n_1 so large that

$$2B \sum_{n > n_1} n^{-\theta} \leq t^p.$$

For each $1 \leq n \leq n_1$, (1) gives a finite-codimensional subspace on which

$$\|x + y\|_n \leq \|x\|_n + K'nt^p$$

whenever $|y| \leq t$, where K' is independent of n, x, t . Intersecting these finitely many subspaces, and then summing, yields

$$\begin{aligned} |x + y| &\leq \sum_{n \leq n_1} n^{-\theta} \|x + y\|_n + \sum_{n > n_1} n^{-\theta} \|x + y\|_n \\ &\leq \sum_{n \leq n_1} n^{-\theta} \|x\|_n + K't^p \sum_{n=1}^{\infty} n^{1-\theta} + \sum_{n > n_1} n^{-\theta} \|x\|_n + t^p \\ &\leq 1 + C_1 t^p. \end{aligned}$$

Here $\sum n^{1-\theta} < \infty$ because $\theta > 2$. Thus $|\cdot|$ is AUS of power type p .

AUC estimate

Let $|x| = 1$ and $0 < t \leq 1$. The AUC assumption for N gives a constant $\eta > 0$ and a finite-codimensional subspace E_0 such that, for all $y \in E_0$ with $|y| \geq t$ and $|y| \leq 1$,

$$N(x + y) \geq N(x)(1 + \eta t^q). \quad (3)$$

Choose a large constant α and an integer n_0 satisfying

$$n_0 - 1 \leq \alpha t^{-q} \leq n_0.$$

Taking α large enough, (2) and (3) imply that, for every $n \geq n_0$,

$$\begin{aligned} \|x + y\|_n &\geq (1 + n^{-1})^{-1} N(x + y) \\ &\geq (1 - n^{-1}) N(x) (1 + \eta t^q) \\ &\geq \|x\|_n + d t^q \end{aligned} \quad (4)$$

with a constant $d > 0$ independent of x, y, t, n . In the last step we used $\|x\|_n \leq N(x)$ and the lower bound $N(x) \geq A|x| = A$.

For the finitely many indices $n < n_0$, choose norming functionals for x in the norms $\|\cdot\|_n$ and intersect their kernels with E_0 . The resulting finite-codimensional subspace E still satisfies (3), and for $y \in E$,

$$\|x + y\|_n \geq \|x\|_n \quad (n < n_0). \quad (5)$$

Combining (4) and (5), for all $y \in E$ with $|y| \geq t$ and $|y| \leq 1$,

$$\begin{aligned} |x + y| &\geq |x| + dt^q \sum_{n \geq n_0} n^{-\theta} \\ &\geq 1 + d_1 t^q n_0^{1-\theta} \\ &\geq 1 + d_2 t^{q\theta}. \end{aligned}$$

Thus $|\cdot|$ is AUC of power type $q\theta$.

Since $q\theta < r$ and $0 < t \leq 1$, the estimate $\bar{\delta}_{|\cdot|}(t) \geq d_2 t^{q\theta}$ implies $\bar{\delta}_{|\cdot|}(t) \geq d_2 t^r$ after decreasing constants if necessary. Therefore $|\cdot|$ is AUC of power type r , while the first part showed it is AUS of power type p . $\square$

Verification Notes

The proof is not computational. It is a direct modification of the source paper’s Asplund averaging construction. The only new point is the tail summation in the AUC estimate. The proof deliberately keeps the same auxiliary dual norms $N^* + n^{-1}M^*$ used in the paper, so that the AUS estimate quoted as (1) is exactly the estimate already used in Lemma 4.3.

The result is a partial answer only. It improves the source-paper exponent $4q$ to every exponent strictly larger than $2q$, but the question from Remark 4.4 asks for the exact exponent q .

Novelty and Search Bounds

Before promoting this packet, the run indexes

`registry_index.tsv`, `solutions/index.tsv`, `attempts/index.tsv`, `proof_gaps/index.tsv`

were searched for arXiv:1506.07978, Rolewicz, AUS/AUC power types, and the phrase “preserve both power types.” The local parsed arXiv source corpus was also searched for the exact phrase and close AUS/AUC power-type formulations. No duplicate packet or local later answer was found.

A bounded web search for close phrases such as “preserve both power types,” “both power types together,” and “simultaneously AUS AUC power type” found no exact later resolution in that search window. This is not a full literature survey, so the novelty confidence should be treated as bounded rather than definitive.

Human Review Recommendation

Recommended verdict after review, if the estimates check out: accept as a partial result improving Lemma 4.3’s quantitative exponent loss. The remaining open question is whether the AUC exponent can be reduced from every $r > 2q$ all the way to q .

References

- [1] S. J. Dilworth, D. Kutzarova, G. Lancien, and L. N. Randrianarivony, *Equivalent norms with the property (β) of Rolewicz*, arXiv:1506.07978, 2015.